Integrated BMS Safety & IoT Monitoring System
Internship Final Integrated Project Report

ESP32 • Wokwi • Modular BMS • Protection Relay • Fault State Machine • LCD • Wi-Fi/Blynk Dashboard

Prepared from the six supplied internship task implementations and integrated into one system.

1. Executive Summary

This project combines the six internship tasks into one embedded safety and monitoring system. The ESP32 reads simulated battery-cell, current and temperature inputs; the scalable BMS engine calculates weakest/strongest cells, imbalance, trend and an adaptive threshold; anomaly detection and a protection relay provide anti-chatter and sensor-integrity protection; a structured fault state machine manages safe-state transitions and recovery; a differential LCD provides local status; and an event-driven telemetry layer sends data to a Blynk dashboard while buffering data during connectivity loss.

2. Integration Objectives

The primary objective is to satisfy the internship requirement that all tasks operate as one complete project rather than six independent submissions. The final architecture therefore has one application loop and shared state flow. Downstream functions consume the BMS snapshot and fault manager state instead of duplicating analysis.

3. Task Integration

Task 1 supplies cell-health analytics. Task 2 consumes current/anomaly information and controls relay protection. Task 3 renders shared state on the LCD without repainting unchanged lines. Task 4 receives isolated faults and controls system-level state. Task 5 owns Wi-Fi reconnection and FIFO telemetry. Task 6 is represented by the dashboard data model and Blynk virtual-pin mapping.

4. Non-Blocking Engineering

The application uses millis()-based scheduling and state machines. Sensor processing, LCD updates, Wi-Fi reconnection, telemetry draining, heartbeat generation and recovery verification are scheduled independently. No blocking delay is used. Telemetry is throttled to avoid bursts during backlog replay.

5. BMS Logic

The BMS engine is scalable through NUM_CELLS. Each cycle finds minimum and maximum valid cell voltage, computes imbalance and its delta, classifies trend, derives an adaptive threshold from average SoC and discharge C-rate, and exposes one BMS_Snapshot for downstream modules.

6. Protection and Anomaly Detection

Sensor integrity checks include hard range validation, frozen-value detection and statistical jump discrimination. A suspicious jump is held for confirmation so a genuine rapid load change can settle without being incorrectly classified as a sensor failure. The relay uses separate trip/clear thresholds, debounce timing, recovery timing and lockout limits.

7. Fault State Machine

Faults are isolated by source and code and retained until explicitly cleared. The state machine supports NORMAL, DEGRADED, FAILSAFE and SHUTDOWN. FAILSAFE recovery is verified through consecutive clean samples and cannot jump directly to NORMAL. Exhausted recovery attempts escalate to SHUTDOWN.

8. LCD / Local HMI

The LCD is updated differentially: a line is written only when its content changes. Three pages present system state, cell health, current/temperature, trend, relay and communications/queue status. Page rotation is timer-driven.

9. IoT Telemetry

Every event enters the same FIFO telemetry path whether the device is online or offline. During an outage, events accumulate; after reconnection, a bounded number of events are replayed per loop tick. Queue depth and

LIVE/REPLAY status are exposed to the dashboard.

10. Dashboard

The intended Blynk dashboard includes cell voltages, weakest/strongest cell, relay status, fault state/count, Wi-Fi RSSI/quality, queue depth, imbalance, SoC, temperature, current, BMS trend, BMS status and anomaly status. The included preview is illustrative; live screenshots must be captured after configuring Blynk.

11. Wokwi Test Controls

Four potentiometers represent cell voltages; additional potentiometers represent current and temperature. Two pushbuttons provide fault and clear actions. The relay output represents the protection actuator.

12. Test Scenarios

Recommended evidence scenarios are normal operation; cell imbalance; high current/relay protection; sensor anomaly; Wi-Fi outage with queue growth; reconnection and replay; and fault recovery. Capture Serial output, LCD state and dashboard state where applicable.

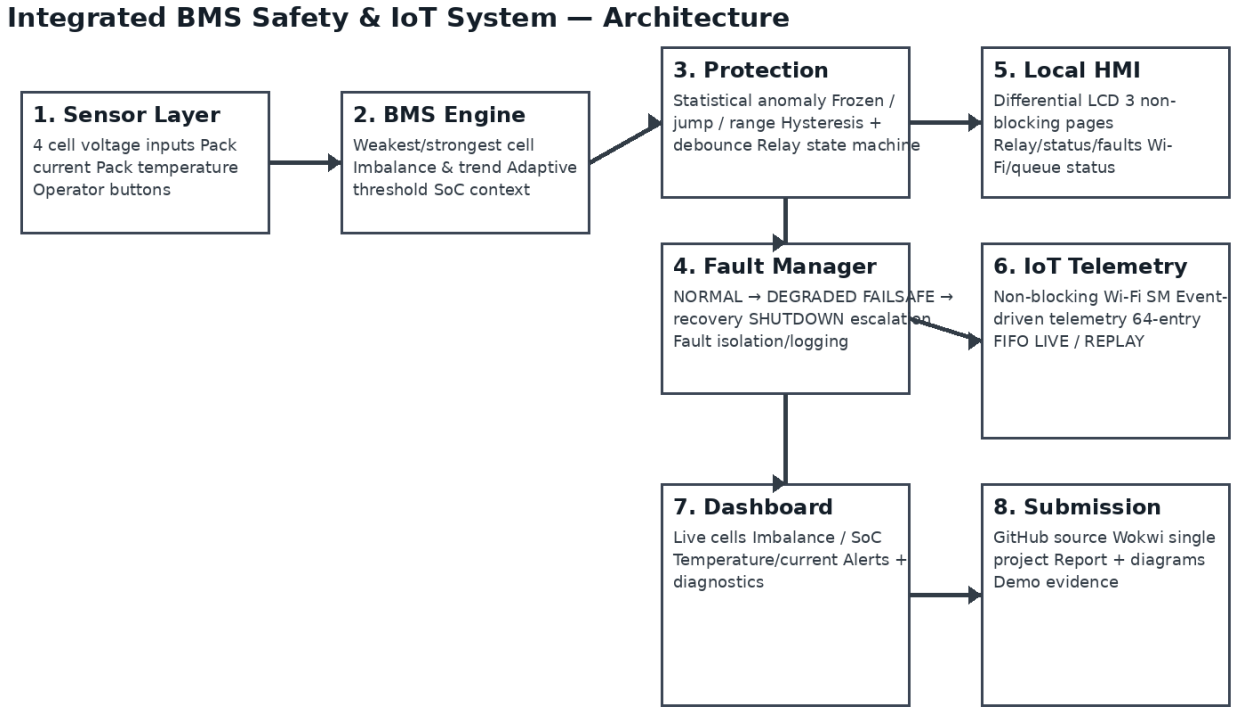
13. Submission Package

The repository contains one integrated firmware project, Wokwi circuit definition, dashboard source, diagrams, reports and a submission checklist. The final external submission still requires the user's GitHub URL, Wokwi URL and live Blynk screenshots/demo recording.

14. Conclusion

The resulting project is structured as a single integrated embedded safety system. Its modular interfaces preserve the original task implementations while providing a shared control loop and fault/telemetry flow. The design addresses modularity, scalability, non-blocking behavior, fault handling, state management and IoT communication.

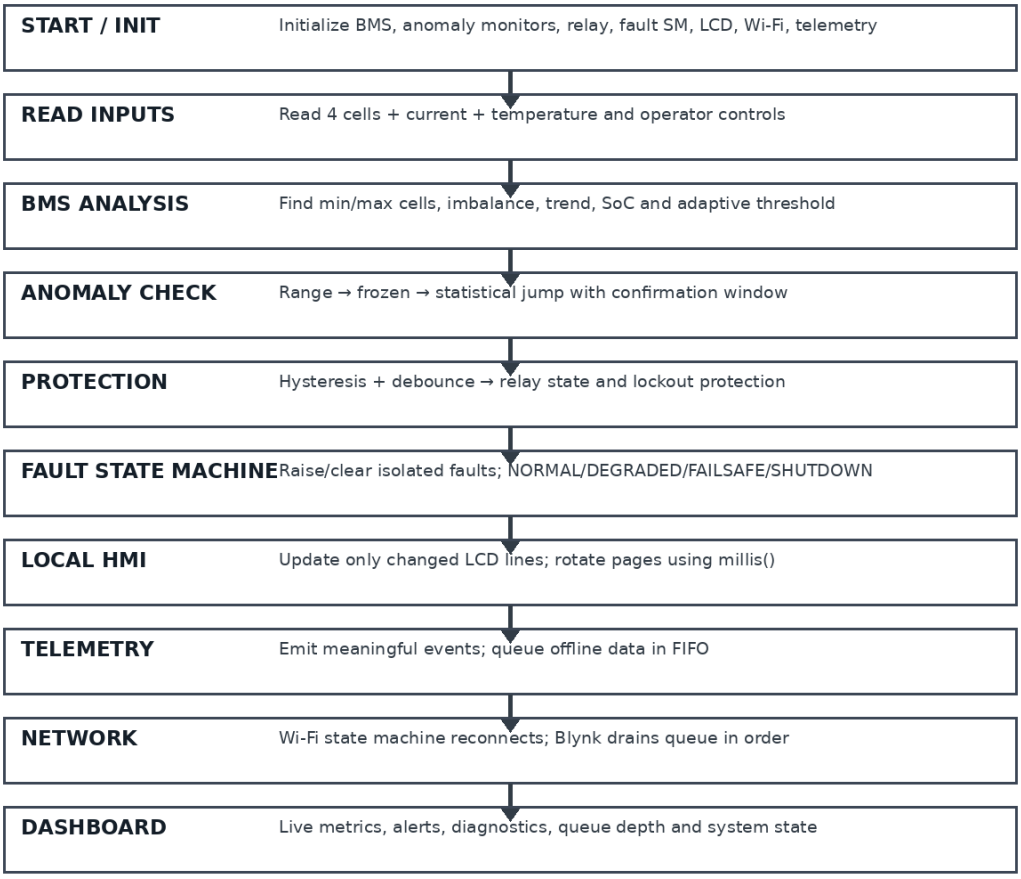
System Architecture Diagram



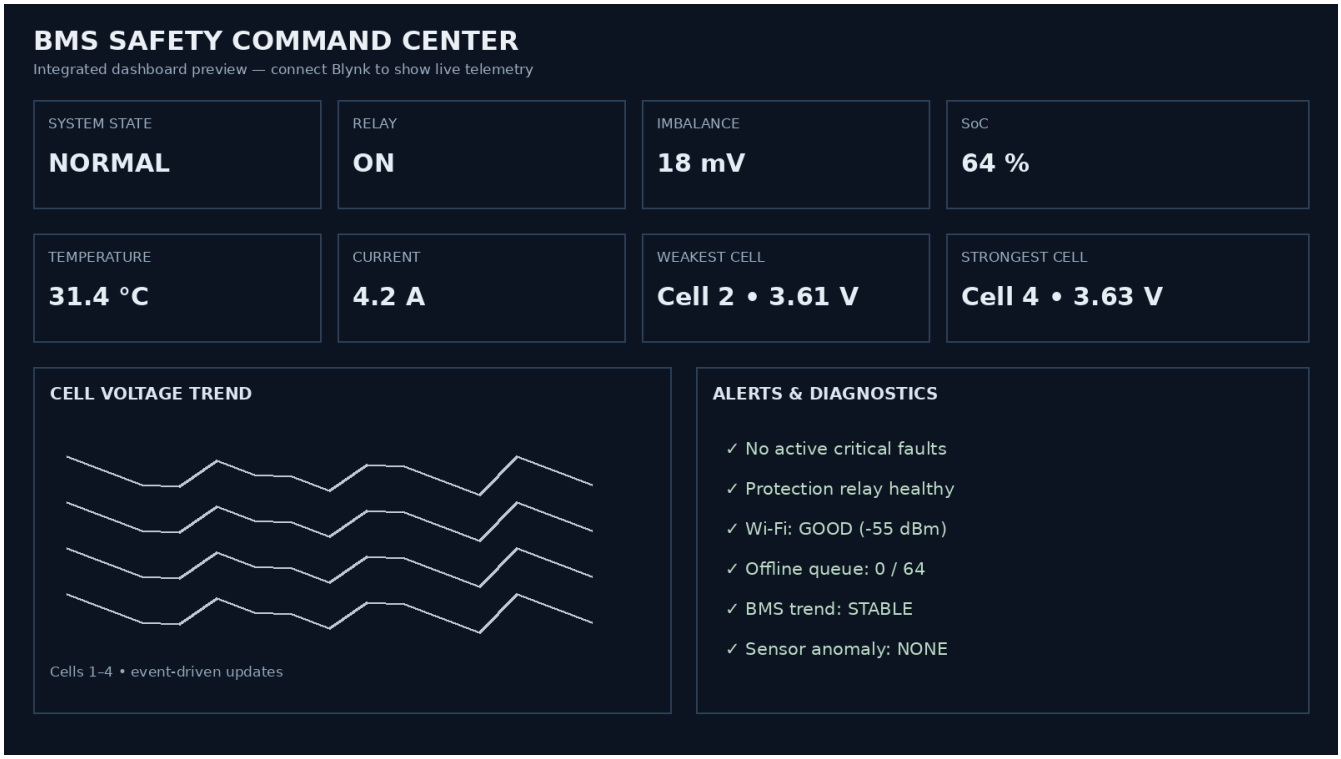
All control paths are driven by millis()/state machines; no blocking delay is used.

Workflow / Block Diagram

Integrated System Workflow / Block Diagram



Dashboard Preview



Important: the dashboard image is an illustrative preview, not a live Blynk screenshot. Replace/add live screenshots before submission.